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BONDING APPARATUS FOR DISPLAY APPARATUS

Abstract

A bonding apparatus for a display apparatus includes a first support unit on which a first panel is mounted and a second support unit on which a second panel is bonded to the first panel is mounted, wherein the second support unit includes a stage on which the second panel is mounted, a main support portion arranged below the stage and supporting the stage, wherein the stage is rotated with the main support portion as a rotation axis as the main support portion is rotated, a sub-support portion connected below the stage and supporting the stage, wherein the sub-support portion slidingly moves on a plane which is perpendicular to the rotation axis as the stage is rotated, and a base plate supporting the main support portion and the sub-support portion.

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Background/Summary

[0001] This application claims priority to Korean Patent Application No. 10-2024-0033320, filed on Mar. 8, 2024, and all the benefits accruing therefrom under 35 USC § 119, the content of which in its entirety is herein incorporated by reference.

BACKGROUND

1. Field

[0002] The invention relates to a bonding apparatus for a display apparatus, and more particularly, to a bonding apparatus for a display apparatus which is capable of bonding panels with increased precision.

2. Description of the Related Art

[0003] Recently, various electronic devices, such as mobile electronic devices and stationary electronic devices, have been used. Such electronic devices include display apparatuses for providing visual information such as an image or a video to a user for supporting various functions.

[0004] Generally, a display apparatus may include a display panel and a display circuit board, where the display panel displays an image and the display circuit board is electrically connected to the display panel and for transmitting signals thereto. The display panel and the display circuit board may be bonded and connected to each other by a bonding apparatus for the display apparatus.

SUMMARY

[0005] To bond a display panel to a display circuit board, the display panel and the display circuit board may be aligned with each other and pressed against each other by a pressing unit. However, in the related art, a support portion that withstands the pressing force of the pressing unit typically has one shaft, and thus, problems such as deflection or pushing of a stage due to eccentricity may occur.

[0006] One or more embodiments include a bonding apparatus for a display apparatus, which is capable of bonding a display panel to a display circuit board with increased precision.

[0007] According to an embodiment, a bonding apparatus for a display apparatus, includes a first support unit on which a first panel is mounted and a second support unit on which a second panel which is bonded to the first panel is mounted, wherein the second support unit includes a stage on which the second panel is mounted, a main support portion arranged below the stage and supporting the stage, wherein the stage is rotated with the main support portion as a rotation axis as the main support portion is rotated, a sub-support portion connected below the stage and supporting the stage, wherein the sub-support portion slidingly moves on a plane which is perpendicular to the rotation axis as the stage is rotated, and a base plate supporting the main support portion and the sub-support portion.

[0008] In an embodiment, the main support portion and the sub-support portion may be arranged to be spaced apart from each other by a first distance, and while the stage is being rotated, the first distance between the main support portion and the sub-support portion may be maintained to be constant.

[0009] In an embodiment and in a plan view, as the stage is rotated, the sub-support portion may linearly move in a first direction and a second direction crossing the first direction on the plane which is perpendicular to the rotation axis.

[0010] In an embodiment, the sub-support portion may include a first linear movement portion arranged on the base plate and linearly moving in the first direction, a connection plate connected above the first linear movement portion, and a second linear movement portion arranged on the connection plate and linearly moving in the second direction.

[0011] In an embodiment and in a plan view, the sub-support portion may perform a curve motion

with respect to the main support portion as the main support portion is rotated.

[0012] In an embodiment, the sub-support portion may include a curvature movement portion arranged on the base plate and having a curved shape and a connection plate connected above the curvature movement portion.

[0013] In an embodiment, the sub-support portion may be provided in plural and may be arranged at each of both sides of the main support portion in a plan view.

[0014] In an embodiment, the sub-support portion may be provided in plural and may be arranged at each of an upper side, a lower side, a right side, and a left side of the main support portion in a plan view.

[0015] In an embodiment, the second support unit may further include a driver driving the main support portion to be rotated, where the driver may not drive the sub-support portion.

[0016] According to an embodiment, a bonding apparatus for a display apparatus includes a first support unit on which a first panel is mounted and a second support unit on which a second panel which is bonded to the first panel is mounted, wherein the second support unit includes a stage on which the second panel is mounted, a main support portion arranged below the stage and supporting the stage, a sub-support portion connected below the stage and arranged to be spaced apart from the main support portion, and a base plate supporting the main support portion and the sub-support portion, wherein as the sub-support portion slidingly moves on a plane which is perpendicular to a rotation axis of the main support portion, the stage connected to the sub-support portion is rotated with respect to the main support portion.

[0017] In an embodiment, the main support portion and the sub-support portion may be arranged to be spaced apart from each other by a first distance, and while the stage is being rotated, the first distance between the main support portion and the sub-support portion may be maintained to be constant.

[0018] In an embodiment and in a plan view, as the sub-support portion linearly moves in a first direction and a second direction crossing the first direction on the plane which is perpendicular to the rotation axis, the stage may be rotated.

[0019] In an embodiment, the sub-support portion may include a first linear movement portion arranged on the base plate and linearly moving in the first direction, a connection plate connected above the first linear movement portion, and a second linear movement portion arranged on the connection plate and linearly moving in the second direction.

[0020] In an embodiment and in a plan view, as the sub-support portion performs a curve motion with respect to the main support portion, the stage may be rotated.

[0021] In an embodiment, the sub-support portion may include a curvature movement portion arranged on the base plate and having a curved shape and a connection plate connected above the curvature movement portion.

[0022] In an embodiment, the sub-support portion may be provided in plural and may be arranged at each of both sides of the main support portion in a plan view.

[0023] In an embodiment, the sub-support portion may be provided in plural and may be arranged at each of an upper side, a lower side, a right side, and a left side of the main support portion in a plan view.

[0024] In an embodiment, the second support unit may further include a driver driving the sub-support portion to linearly move the sub-support portion, where the driver may not drive the main support portion.

[0025] According to an embodiment, a method of manufacturing a display apparatus comprises arranging a first panel on a first support unit, arranging a second panel on a second support unit, correcting a shaft rotation of the second support unit for the alignment of the second panel, and pressing the first panel above the second support unit to bond the first panel to the second panel while a portion of the first panel is disposed on the second support unit to overlap the second panel, wherein the second support unit comprises a stage on which the second panel is mounted, a main

support portion arranged below the stage, and a sub-support portion connected below the stage, wherein the correcting of the shaft rotation comprises a step in which the stage is rotated with the main support portion as a rotation axis as the main support portion is rotated, and the sub-support portion slidingly moves on a plane directed perpendicular to the rotation axis as the stage rotated.

[0026] According to an embodiment, a method of manufacturing a electronic device comprises manufacturing a display apparatus and accommodating the display apparatus in the housing, wherein manufacturing the display apparatus comprises arranging a first panel on a first support unit, arranging a second panel on a second support unit, correcting a shaft rotation of the second support unit for the alignment of the second panel, and pressing the first panel above the second support unit to bond the first panel to the second panel while a portion of the first panel is disposed on the second support unit to overlap the second panel, wherein the second support unit comprises a stage on which the second panel is mounted, a main support portion arranged below the stage, and a sub-support portion connected below the stage, wherein the correcting of the shaft rotation comprises a step in which the stage is rotated with the main support portion as a rotation axis as the main support portion is rotated, and the sub-support portion slidingly moves on a plane directed perpendicular to the rotation axis as the stage rotated.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] The above and other aspects, features, and advantages of certain embodiments of the invention will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0028] FIG. 1 is a schematic plan view of a portion of a display apparatus showing a display panel in an unbent state, according to an embodiment;

[0029] FIG. 2 is a schematic lateral cross-sectional view of a portion of a display apparatus showing a display panel in a bent state, according to an embodiment;

[0030] FIG. 3 is a schematic equivalent circuit diagram of a pixel circuit applicable to a display apparatus, according to an embodiment;

[0031] FIG. 4 is a schematic cross-sectional view of a portion of a display apparatus of FIG. 1 taken along line A-A', according to an embodiment;

[0032] FIG. 5 is a schematic cross-sectional view of a bonding apparatus for a display apparatus, according to an embodiment;

[0033] FIG. 6 is a schematic front view of a second support unit, wherein a first support unit is omitted, according to an embodiment;

[0034] FIG. 7 is a schematic plan view of a second support unit illustrating a main support portion and a sub-support portion, wherein a first support unit is omitted, according to an embodiment; and

[0035] FIG. 8 is a schematic plan view of a second support unit which is substantially the same as FIG. 7, according to an embodiment.

DETAILED DESCRIPTION

[0036] Reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings, wherein like reference numerals refer to like elements throughout. In this regard, the present embodiments may have different forms and should not be construed as being limited to the descriptions set forth herein. Accordingly, the embodiments are merely described below, by referring to the figures, to explain aspects of the present description. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Throughout the disclosure, the expression “at least one of a, b or c” indicates only a, only b, only c, both a and b, both a and c, both b and c, all of a, b, and c, or variations thereof.

[0037] While the invention is capable of various modifications and alternative forms, embodiments

thereof are shown by way of example in the drawings and will herein be described in detail. Effects and characteristics of the invention and methods of achieving the same will become apparent by referring to the embodiments described in detail below along with the drawings. However, the invention is not limited to the embodiments disclosed hereinafter and may be realized in various forms.

[0038] Hereinafter, embodiments will be described in detail by referring to the accompanying drawings, wherein, when describing the accompanying drawings, elements that are the same as or corresponding to each other will be assigned the same reference numerals, repeated descriptions thereof will not be given.

[0039] It will be understood that although the terms “first,” “second,” etc. may be used herein to describe various components, these components should not be limited by these terms. These components are only used to distinguish one component from another.

[0040] As used herein, the singular expressions “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.

[0041] It will be further understood that the terms “comprises” and/or “comprising” used herein specify the presence of stated features or elements, but do not preclude the presence or addition of one or more other features or elements.

[0042] It will be further understood that when a layer, region, or element is referred to as being formed “on” another layer, area, or element, it can be directly or indirectly formed on the other layer, region, or element. That is, for example, intervening layers, regions, or elements may be present.

[0043] Sizes of elements in the drawings may be exaggerated for convenience of explanation. For example, sizes and thicknesses of the elements in the drawings are randomly indicated for convenience of explanation, and thus, the invention is not necessarily limited to the illustrations of the drawings.

[0044] In the embodiments hereinafter, the x-axis, the y-axis and the z-axis are not limited to three axes of the rectangular coordinate system and may be interpreted in a broader sense. For example, the x-axis, the y-axis, and the z-axis may be perpendicular to one another, or may represent different directions that are not perpendicular to one another.

[0045] When a certain embodiment may be implemented differently, a specific process order may be performed differently from the described order. For example, two consecutively described processes may be performed substantially at the same time or performed in an order opposite to the described order.

[0046] FIG. 1 is a schematic plan view of a portion of a display apparatus 1, according to an embodiment, and shows a display panel 10 in an unbent state for convenience of explanation. FIG. 2 is a schematic lateral cross-sectional view of a portion of the display apparatus 1, according to an embodiment, and shows the display panel 10 in a bent state.

[0047] In an embodiment and referring to FIGS. 1 and 2, the display apparatus 1 may display a motion image or a static image and may be used as a display screen not only of portable electronic devices, such as a mobile phone, a smartphone, a tablet personal computer (PC), a mobile communication terminal, an electronic notebook, an electronic book, a portable multimedia player (PMP), a navigation device, and an ultra-mobile PC (UMPC), but also of various products, such as a television (TV), a notebook computer, a monitor, a signboard, an Internet of things (IOT) device, etc. Also, the display apparatus 1, according to an embodiment, may be used for wearable devices, such as a smart watch, a watch phone, a glasses-type display, and a head-mounted display (HMD). Also, the display apparatus 1, according to an embodiment, may be used as a center information display (CID) on a gauge of a vehicle or a center fascia or a dashboard of a vehicle, a room mirror display substituting a side-view mirror of a vehicle or a display arranged on a rear surface of a front seat, as an entertainment device for a backseat of a vehicle.

[0048] In an embodiment, the display apparatus 1 may be accommodated in a housing of the

electronic device. The housing may serve as a cover that protects internal components such as the display apparatus **1** and forms an exterior of the electronic device. In addition, the display apparatus **1** may be connected to an electronic module of the electronic device and may be driven on the electronic device. Hereinafter, the explanation will focus on the display apparatus **1**.

[0049] In an embodiment and as illustrated in FIG. **1**, the display apparatus **1** may approximately have a rectangular shape. For example, as illustrated in FIG. **1**, the display apparatus **1** may generally have a rectangular planar shape having a short side extending in a first direction (for example, an x direction or a -x direction) and a long side extending in a second direction (for example, a y direction or a -y direction). According to an embodiment, a portion of the display apparatus **1**, at which the short side extending in the first direction (for example, the x direction or the -x direction) and the long side extending in the second direction (for example, the y direction or the -y direction) meet each other, may have a rectangular shape or a circular shape having a certain curvature. However, the display apparatus **1** is not limited to the rectangular planar shape and may include other planar shapes, such as a polygonal planar shape, a circular planar shape, or an elliptical planar shape.

[0050] The display apparatus **1** may include a display area DA and a peripheral area PA, where the display area DA may display an image. Here, pixels PX may be arranged in the display area DA. The display apparatus **1** may provide an image by using light emitted from the pixels PX, where each pixel PX may emit light by using a display element. According to an embodiment, each pixel PX may emit red, green, or blue light. According to another embodiment, each pixel PX may emit red, green, blue, or white light.

[0051] In an embodiment, the peripheral area PA may be a non-display area which does not display an image. The peripheral area PA may at least partially surround the display area DA. For example, the peripheral area PA may entirely surround the display area DA. A driver configured to provide electrical signals to the pixels PX and power lines configured to provide a power supply, or the like may be arranged in the peripheral area PA. For example, a scan driver configured to apply scan signals to the pixels PX may be arranged in the peripheral area PA. Also, a data driver configured to apply data signals to the pixels PX may be arranged in the peripheral area PA. According to an embodiment, the peripheral area PA may include an adjacent area AA disposed adjacent to the display area DA and surrounding the display area DA, a bending area BA connected to a side of the adjacent area AA and capable of being bent, and a pad area PDA to which the bending area BA is connected and in which pads are arranged.

[0052] In an embodiment and referring to FIG. **2**, the display apparatus **1** may include the display panel **10**, a cover window **20**, a display driver **30**, a display circuit board **40**, a touch sensor driver **50**, a cover panel **60**, and a protective film PTF.

[0053] The display panel **10** may display information processed by the display apparatus **1**. For example, the display panel **10** may display execution screen information of an application driven by the display apparatus **1** or the display panel **10** may display user interface (UI) or graphics user interface (GUI) information based on the execution screen information.

[0054] The display panel **10** may include a display element. For example, the display panel **10** may include an organic light-emitting display panel using an organic light-emitting diode, a micro light-emitting diode display panel using a micro light-emitting diode, a quantum-dot light-emitting display panel using a quantum-dot light-emitting diode including a quantum-dot emission layer, or an inorganic light-emitting display panel using an inorganic light-emitting diode including an inorganic semiconductor. Hereinafter, an embodiment where the display panel **10** includes an organic light-emitting display panel using an organic light-emitting diode as a display element will be mainly described in detail.

[0055] In an embodiment, the display panel **10** may include a substrate **100** and multiple layers disposed on the substrate **100**. According to an embodiment, the display panel **10** may include the substrate **100**, a display layer DSL, an encapsulation layer TFE, a touch sensor layer TSL, and an

optical functional layer OFL. Here, the display area DA and the peripheral area PA may be defined in the substrate **100** and/or the multiple layers. For example, the substrate **100** may include the display area DA and the peripheral area PA. Also, the peripheral area PA may include the pad area PDA and the bending area BA.

[0056] In an embodiment, the substrate **100** may include polymer resins, such as polyethersulfone, polyarylate, polyetherimide, polyethylene naphthalate, polyethylene terephthalate, polyphenylene sulfide, polyimide, polycarbonate, cellulose tri acetate, or cellulose acetate propionate. According to an embodiment, the substrate **100** may have a multi-layered structure including a base layer including the polymer resins described above and a barrier layer (not shown). The substrate **100** including the polymer resins may be flexible, rollable, or bendable.

[0057] In an embodiment, the substrate **100** may be bent in the bending area BA. In this case, according to an embodiment, the substrate **100** may be bent in the bending area BA to have a “U” shape, so that at least portions of a lower surface **100LS** of the substrate **100** may face each other, and the pad area PDA of the substrate **100** may be positioned below other portions of the substrate **100**. Thus, an area of the peripheral area PA, the area which is visible to a user, may be reduced. FIG. 2 illustrates that only the substrate **100** is bent. However, according to another embodiment, at least a portion of the display layer DSL, at least a portion of the encapsulation layer TFE, and at least a portion of the touch sensor layer TSL may also correspond to the bending area BA and the pad area PDA. In this case, the at least the portion of the display layer DSL, the at least the portion of the encapsulation layer TFE, and the at least the portion of the touch sensor layer TSL may also be bent in the bending area BA.

[0058] In an embodiment, the display layer DSL may be disposed on the substrate **100** and may include pixel circuits and display elements. Here, the pixel circuits may be connected to the display elements, respectively. The pixel circuit may include a transistor and a storage capacitor. Thus, the display layer DSL may include the plurality of display elements, a plurality of transistors, and storage capacitors. Also, the display layer DSL may further include insulating layers which are disposed between the plurality of display elements, the plurality of transistors, and the storage capacitors.

[0059] In an embodiment, the encapsulation layer TFE may be disposed on the display layer DSL, where the encapsulation layer TFE may be disposed on the display element and may cover the display element. According to an embodiment, the encapsulation layer TFE may include at least one inorganic encapsulation layer and at least one organic encapsulation layer. The at least one inorganic encapsulation layer may include one or more inorganic materials from among Al.sub.2O.sub.3, TiO.sub.2, Ta.sub.2O.sub.5, ZnO, SiO.sub.2, SiN.sub.x, and SiON. The at least one organic encapsulation layer may include a polymer-based material. The polymer-based material may include acryl-based resins, epoxy-based resins, polyimide, polyethylene, etc.

According to an embodiment, the at least one organic encapsulation layer may include acrylate.

[0060] In an embodiment, the touch sensor layer TSL may be disposed on the encapsulation layer TFE, where the touch sensor layer TSL may sense coordinate information according an external input, for example, a touch event. The touch sensor layer TSL may include a sensor electrode and touch lines connected to the sensor electrode. The touch sensor layer TSL may sense an external input according to a magnetic capacitance method or a mutual capacitance method.

[0061] In an embodiment, the touch sensor layer TSL may be formed on the encapsulation layer TFE. In another embodiment, after the touch sensor layer TSL may be separately formed on a touch substrate, the touch sensor layer TSL may be attached onto the encapsulation layer TFE through an adhesive layer, such as an optically clear adhesive. According to an embodiment, the touch screen layer TSL may be formed directly on the encapsulation layer TFE, and in this case, the adhesive layer may not be disposed between the touch sensor layer TSL and the encapsulation layer TFE.

[0062] In an embodiment, the optical functional layer OFL may be disposed on the touch sensor layer TSL and may reduce the reflectivity of light (external light) which may be incident from the

outside toward the display apparatus **1** and/or improve the color purity of light which is emitted from the display apparatus **1**. According to an embodiment, the optical functional layer OFL may include a phase retarder and a polarizer. The phase retarder may include a film-type phase retarder or a liquid crystal coating-type phase retarder, and may include a $\lambda/2$ phase retarder and/or a $\lambda/4$ phase retarder. The polarizer may also include a film-type polarizer or a liquid crystal coating-type polarizer. The film-type polarizer may include an elongation-type synthetic resin film, and the liquid crystal coating-type polarizer may include liquid crystals arranged in a certain shape. The phase retarder and the polarizer may further include a protective film.

[0063] According to another embodiment, the optical functional layer OFL may include a black matrix and color filters, where the color filters may be arranged by taking into account a color of the light emitted from each pixel in the display apparatus **1**. Each color filter may include a red, green, or blue pigment or dye. In another embodiment, each color filter may further include quantum dots, in addition to the pigment or the dye described above. In still another embodiment, some of the color filters may not include the pigment or the dye described above and may include scattered particles, such as oxide titanium.

[0064] According to another embodiment, the optical functional layer OFL may include a destructive interference structure, where the destructive interference structure may include a first reflective layer and a second reflective layer disposed on different layers. First reflective light and second reflective light reflected from the first reflective layer and the second reflective layer, respectively, may destructively interfere, and thus, the reflectivity of external light may be decreased.

[0065] In an embodiment, the cover window **20** may be disposed on the display panel **10** and may protect the display panel **10**. According to an embodiment, the cover window **20** may include a flexible window. The cover window **20** may protect the display panel **10** by being easily bent through an external force without the occurrence of cracks, etc. The cover window **20** may include at least one of glass, sapphire, and plastic. The cover window **20** may include, for example, ultra-thin glass (UTG) or colorless polyimide (CPI). According to an embodiment, the cover window **20** may have a structure in which a flexible polymer layer is disposed on a surface of a glass substrate or may have a structure including only a polymer layer.

[0066] In an embodiment, the cover window **20** may be attached onto the display panel **10** through an adhesive member, where the adhesive member may include a transparent adhesive member such as an optically clear adhesive (OCA). In addition, the adhesive member may include various other well-known adhesive materials. The adhesive member may be attached above the display panel **10** as a type of film, or a type of material may be spread above the display panel **10** as the adhesive member. As such, the adhesive member may be formed above the display panel **10** in various ways.

[0067] In an embodiment, the display driver **30** may be arranged in the pad area PDA and may receive control signals and power voltages and may generate and output signals and voltages for driving the display panel **10**. The display driver **30** may include an integrated circuit (IC).

[0068] In an embodiment, the display circuit board **40** may be electrically connected to the display panel **10**. For example, the display circuit board **40** may be electrically connected to the pad area PDA of the substrate **100** by an anisotropic conductive film.

[0069] The display circuit board **40** may include a flexible printed circuit board (FPCB) which may be a bent or a rigid printed circuit board (PCB) which is rigid and not easily bent. In another embodiment, the display circuit board **40** may include a complex printed circuit board including both a PCB and an FPCB.

[0070] In an embodiment, the touch sensor driver **50** may be disposed on the display circuit board **40** and may include an IC. The touch sensor driver **50** may be attached to the display circuit board **40**. The touch sensor driver **50** may be electrically connected to sensor electrodes of the touch sensor layer TSL of the display panel **10** through the display circuit board **40**.

[0071] However, in addition thereto, a power supply portion may further be disposed on the display

circuit board **40**, where the power supply portion may supply a driving voltage for driving the pixels of the display panel **10** and the display driver **30**.

[0072] In an embodiment, the protective film PTF may be patterned and attached onto the lower surface **100LS** of the substrate **100**, where the protective film PTF may be attached onto areas of the substrate **100** with the exception of the bending area BA. Here, a portion of the protective film PTF may be disposed on the opposite side of the display layer DSL with the substrate **100** disposed therebetween. Another portion of the protective film PTF may be attached onto the lower surface **100LS** of the substrate **100** to correspond to the pad area PDA.

[0073] According to an embodiment, the cover panel **60** may be disposed between the portions of the protective film PTF. In other words, the cover panel **60** may be disposed between a portion (for example, a first portion) and another portion (for example, a second portion) of the substrate **100** that face each other as the substrate **100** is bent to have the “U” shape. The cover panel **60** may absorb external shocks to prevent the fracture of the display panel **10**.

[0074] FIG. **3** is a schematic equivalent circuit diagram of a pixel circuit PC applicable to a display apparatus, according to an embodiment.

[0075] In an embodiment and referring to FIG. **3**, the pixel circuit PC may be electrically connected to a display element DE and may include a first transistor T1, a second transistor T2, and a storage capacitor Cst. According to an embodiment, the display element DE may emit red, green, or blue light or red, green, blue, or white light.

[0076] In an embodiment, the second transistor T2 may be connected to a scan line SL and a data line DL and may be configured to transmit, to the first transistor T1, a data signal or a data voltage which is input from the data line DL, in response to a scan signal or a switching voltage which is input from the scan line SL.

[0077] In an embodiment, the storage capacitor Cst may be connected to the second transistor T2 and a driving voltage line PL and may be configured to store a voltage corresponding to a difference between a voltage received from the second transistor T2 and a first power voltage ELVDD supplied to the driving voltage line PL.

[0078] In an embodiment, the first transistor T1 may be connected to the driving voltage line PL and the storage capacitor Cst and may be configured to control a driving current flowing from the driving voltage line PL through the display element DE according to a value of the voltage stored in the storage capacitor Cst. The display element DE may emit light having a certain brightness according to the driving current. An opposite electrode of the display element DE may receive a second power voltage ELVSS.

[0079] FIG. **3** illustrates that the pixel circuit PC may include two transistors and one storage capacitor. However, in other embodiments, the pixel circuit PC may include more transistors and/or more storage capacitors than the transistors and/or the storage capacitors illustrated in FIG. **3**.

[0080] FIG. **4** is a schematic cross-sectional view of a portion of the display apparatus **1** of FIG. **1** taken along line A-A' in FIG. **1**, according to an embodiment.

[0081] In an embodiment and referring to FIG. **4**, the display apparatus **1** may include the display panel **10**. The display panel **10** may include the substrate **100**, the display layer DSL, an encapsulation layer **300**, the touch sensor layer TSL, and the optical functional layer OFL. For convenience of explanation, in the illustration of FIG. **4**, the touch sensor layer TSL and the optical functional layer OFL are omitted.

[0082] In an embodiment, the display layer DSL may be disposed on the substrate **100** and may include a buffer layer **111**, a pixel circuit layer PCL, and a display element layer DEL.

[0083] The substrate **100** may include glass or may include polymer resins such as polyethersulfone, polyarylate, polyetherimide, polyethylene naphthalate, polyethylene terephthalate, polyphenylene sulfide, polyimide, polycarbonate, cellulose tri acetate, or cellulose acetate propionate. The substrate **100** including the polymer resins may be flexible, rollable, or bendable. The substrate **100** may have a multi-layered structure including a base layer including the

polymer resins described above and a barrier layer (not shown).

[0084] In an embodiment, the buffer layer **111** may include an inorganic insulating material, such as SiN.sub.x, SiON, and SiO.sub.2, and may include a single layer or layers including the inorganic insulating material described above.

[0085] In an embodiment, the pixel circuit layer PCL may be disposed on the buffer layer **111** and may include a transistor TFT included in a pixel circuit, and an inorganic insulating layer IIL, a first planarization layer **115**, and a second planarization layer **116** disposed below or/and above components of the transistor TFT. The inorganic insulating layer IIL may include a first gate insulating layer **112**, a second gate insulating layer **113**, and an interlayer insulating layer **114**.

[0086] In an embodiment, the transistor TFT may include a semiconductor layer A, and the semiconductor layer A may include polysilicon. In another embodiment, the semiconductor layer A may include amorphous silicon, an oxide semiconductor, or an organic semiconductor. The semiconductor layer A may include a channel area, a drain area and a source area arranged at both sides of the channel area, respectively. A gate electrode G may overlap the channel area.

[0087] In an embodiment, the gate electrode G may include a low-resistance metal material. The gate electrode G may include a conductive material including Mo, Al, Cu, Ti, etc. and may include layers or a single layer including the conductive material described above.

[0088] In an embodiment, the first gate insulating layer **112** disposed between the semiconductor layer A and the gate electrode G may include an inorganic insulating material, such as SiO.sub.2, SiN.sub.x, SiON, Al.sub.2O.sub.3, TiO.sub.2, Ta.sub.2O.sub.5, HfO.sub.2, ZnO.sub.x, or the like. ZnO.sub.x may include ZnO and/or ZnO.sub.2.

[0089] In an embodiment, the second gate insulating layer **113** may be provided to cover the gate electrode G. Similarly to the first gate insulating layer **112**, the second gate insulating layer **113** may include an inorganic insulating material, such as SiO.sub.2, SiN.sub.x, SiON, Al.sub.2O.sub.3, TiO.sub.2, Ta.sub.2O.sub.5, HfO.sub.2, ZnO.sub.x, or the like. ZnO.sub.x may include ZnO and/or ZnO.sub.2.

[0090] In an embodiment, an upper electrode CE2 of a storage capacitor Cst may be disposed above the second gate insulating layer **113**, where the upper electrode CE2 may overlap the gate electrode G there-below. Here, the gate electrode G and the upper electrode CE2 overlapping each other with the second gate insulating layer **113** disposed therebetween may form the storage capacitor Cst of the pixel circuit. That is, the gate electrode G may function as a lower electrode CE1 of the storage capacitor Cst. As described above, the storage capacitor Cst and the transistor TFT may be formed to overlap each other. According to some embodiments, the storage capacitor Cst may not be formed to overlap the transistor TFT.

[0091] In an embodiment, the upper electrode CE2 may include Al, Pt, Pd, Ag, Mg, Au, Ni, Nd, Ir, Cr, Ca, Mo, Ti, W, and/or Cu and may include a single layer or layers including the material described above.

[0092] In an embodiment, the interlayer insulating layer **114** may cover the upper electrode CE2. The interlayer insulating layer **114** may include SiO.sub.2, SiN.sub.x, SiON, Al.sub.2O.sub.3, TiO.sub.2, Ta.sub.2O.sub.5, HfO.sub.2, ZnO.sub.x, or the like. ZnO.sub.x may include ZnO and/or ZnO.sub.2. The interlayer insulating layer **114** may include a single layer or layers including the inorganic insulating material described above.

[0093] In an embodiment, each of a drain electrode D and a source electrode S may be disposed on the interlayer insulating layer **114**, where the drain electrode D and the source electrode S may include a highly conductive material. The drain electrode D and the source electrode S may include a conductive material including Mo, Al, Cu, Ti, etc. and may include layers or a single layer including the material described above. According to an embodiment, the drain electrode D and the source electrode S may have a layered structure of Ti/Al/Ti layers.

[0094] In an embodiment, the first planarization layer **115** may be disposed to cover the drain electrode D and the source electrode S, where the first planarization layer **115** may include an

organic insulating layer. The first planarization layer **115** may include an organic insulating material, such as a general-purpose polymer such as polymethylmethacrylate (PMMA) or polystyrene (PS), a polymer derivative having a phenol-based group, an acryl-based polymer, an imide-based polymer, an aryl ether-based polymer, an amide-based polymer, a fluorine-based polymer, a p-xylene-based polymer, a vinyl alcohol-based polymer, and a blend thereof.

[0095] In an embodiment, a connection electrode CML may be disposed on the first planarization layer **115**. Here, the connection electrode CML may be connected to the drain electrode D or the source electrode S through a contact hole of the first planarization layer **115**. The connection electrode CML may include a highly conductive material. The connection electrode CML may include a conductive material including Mo, Al, Cu, Ti, etc. and may include layers or a single layer including the conductive material described above. According to an embodiment, the connection electrode CML may have a layered structure of Ti/Al/Ti.

[0096] In an embodiment, the second planarization layer **116** may be disposed to cover the connection electrode CML. The second planarization layer **116** may include an organic insulating layer. The second planarization layer **116** may include an organic insulating material, such as a general-purpose polymer such as PMMA or PS, a polymer derivative having a phenol-based group, an acryl-based polymer, an imide-based polymer, an aryl ether-based polymer, an amide-based polymer, a fluorine-based polymer, a p-xylene-based polymer, a vinyl alcohol-based polymer, and a blend thereof.

[0097] In an embodiment, the display element layer DEL may be disposed on the pixel circuit layer PCL and may include a display element DE. The display element DE may include an organic light-emitting diode OLED. A pixel electrode **211** of the display element DE may be electrically connected to the connection electrode CML through a contact hole of the second planarization layer **116**.

[0098] In an embodiment, the pixel electrode **211** may include conductive oxide, such as indium tin oxide (ITO), indium zinc oxide (IZO), zinc oxide (ZnO), indium oxide (In.sub.2O.sub.3), indium gallium oxide (IGO), or aluminum zinc oxide (AZO). According to another embodiment, the pixel electrode **211** may include a reflective layer including Ag, Mg, Al, Pt, Pd, Au, Ni, Nd, Ir, Cr, or a compound thereof. According to still another embodiment, the pixel electrode **211** may further include a layer including ITO, IZO, ZnO, or In.sub.2O.sub.3 above/below the reflective layer described above.

[0099] In an embodiment, a pixel-defining layer **118** having an opening **118OP** exposing a central portion of the pixel electrode **211** may be disposed on the pixel electrode **211**. The pixel-defining layer **118** may include an organic insulating material and/or an inorganic insulating material. The opening **118OP** may define an emission area (hereinafter, referred to as an emission area EA) of light emitted from the display element DE. For example, a width of the opening **118OP** may correspond to a width of the emission area EA of the display element DE.

[0100] In an embodiment, a spacer **119** may be disposed on the pixel-defining layer **118**, where the spacer **119** may be provided to prevent the fracture of the substrate **100** according to a method of manufacturing the display apparatus. In a manufacturing process of the display panel, a mask sheet may be used. Here, a portion of the substrate **100** may be damaged or fractured due to the mask sheet, which is inserted into the opening **118OP** of the pixel-defining layer **118** or which is made to adhere to the pixel-defining layer **118** when a deposition material is deposited on the substrate **100**. The spacer **119** may prevent this damage or fracture.

[0101] In an embodiment, the spacer **119** may include an organic insulating material, such as polyimide. In another embodiment, the spacer **119** may include an inorganic insulating material, such as SiNx or SiO₂, or may include an organic insulating material and an inorganic insulating material.

[0102] According to an embodiment, the spacer **119** may include a material which is different from the material of the pixel-defining layer **118**. In another embodiment, the spacer **119** may include a

material of which is the same as the material of the pixel-defining layer **118**, and in this case, the pixel-defining layer **118** and the spacer **119** may be formed together by a mask process using a halftone mask, etc.

[0103] In an embodiment, an intermediate layer **212** may be disposed on the pixel-defining layer **118**, where the intermediate layer **212** may include an emission layer **212b** disposed in the opening **118OP** of the pixel-defining layer **118**. The emission layer **212b** may include a high molecular-weight or a low molecular-weight organic material emitting a certain color of light.

[0104] In an embodiment, a first functional layer **212a** and a second functional layer **212c** may be disposed below and above the emission layer **212b**, respectively. The first functional layer **212a** may include, for example, a hole transport layer (HTL), or an HTL and a hole injection layer (HIL). The second functional layer **212c** may be arranged above the emission layer **212b** and may be optionally arranged. The second functional layer **212c** may include an electron transport layer (ETL) and/or an electron injection layer (EIL). The first functional layer **212a** and/or the second functional layer **212c** may be a common layer formed to entirely cover the substrate **100**, like an opposite electrode **213** to be described below.

[0105] In an embodiment, the opposite electrode **213** may include a conductive material having a low work function. For example, the opposite electrode **213** may include a (semi) transparent layer including Ag, Mg, Al, Pt, Pd, Au, Ni, Nd, Ir, Cr, Li, Ca, or an alloy thereof. In another embodiment, the opposite electrode **213** may further include a layer including ITO, IZO, ZnO, or In.sub.2O.sub.3 on the (semi) transparent layer including the material described above.

[0106] According to some embodiments, a capping layer (not shown) may further be disposed on the opposite electrode **213**, where the capping layer may include LiF, an inorganic material, or/and an organic material.

[0107] In an embodiment, the encapsulation layer **300** may be disposed on the opposite electrode **213**. The encapsulation layer **300** may be disposed on the display element layer DEL and may cover the display element layer DEL. The encapsulation layer **300** may include at least one inorganic encapsulation layer and at least one organic encapsulation layer. According to an embodiment, FIG. 4 illustrates that the encapsulation layer **300** may include a first inorganic encapsulation layer **310**, an organic encapsulation layer **320**, and a second inorganic encapsulation layer **330** which are sequentially stacked.

[0108] In an embodiment, the first inorganic encapsulation layer **310** and the second inorganic encapsulation layer **330** may include at least one inorganic material from among aluminum oxide, titanium oxide, tantalum oxide, hafnium oxide, ZnO, SiOx, SiNx, and SiON. The organic encapsulation layer **320** may include a polymer-based material. The polymer-based material may include acryl-based resins, epoxy-based resins, polyimide, polyethylene, etc. According to an embodiment, the organic encapsulation layer **320** may include acrylate. The organic encapsulation layer **320** may be formed by curing a monomer or by being coated with a polymer. The organic encapsulation layer **320** may be transparent.

[0109] Although not shown, in an embodiment, the touch sensor layer described above may be disposed on the encapsulation layer **300**, and the optical functional layer may be disposed on the touch sensor layer.

[0110] FIG. 5 is a schematic cross-sectional view of a bonding apparatus **2** for the display apparatus **1**, according to an embodiment. According to an embodiment, the bonding apparatus **2** for the display apparatus **1** may be used to bond the components of the display apparatus **1** described above. FIG. 6 is a schematic front view of a second support unit **500**, according to an embodiment. In FIG. 6, a first support unit **400** is omitted, for convenience of explanation.

[0111] In an embodiment and referring to FIG. 5, the bonding apparatus **2** for the display apparatus **1** may be used to bond the components of the display apparatus **1**, for example, the display panel **10** and the display circuit board **40**. According to an embodiment, the bonding apparatus **2** for the display apparatus **1** may include the first support unit **400** and the second support unit **500**.

[0112] In an embodiment, the first support unit **400** may support a first panel and the second support unit **500** may support a second panel. Also, to bond the first panel and the second panel to each other, a portion of the first panel may be disposed on the second support unit **500** to overlap the second panel. Next, a pressing unit (not shown) may press the first panel above the second support unit **500** to bond the first panel to the second panel. Here, a pressing weight may be mostly concentrated in the second support unit **500**. Hereinafter, the first support unit **400** and the second support unit **500** will be described in detail.

[0113] In an embodiment, the first support unit **400** may support the first panel, for example, the display panel **10**. Hereinafter, an embodiment where the first panel is the display panel **10** is mainly described. The display panel **10** may be mounted on a first stage **410**. According to an embodiment, the first stage **410** may have a quadrangular cross-section having a long side extending in a first direction (for example, an x direction of FIG. 5) and a short side extending in a second direction (for example, a y direction of FIG. 5) crossing the first direction. However, the first stage **410** may have the short side extending in the first direction and the long side extending in the second direction or may have a square cross-section having the sides having the same length. Accordingly, the first stage **410** may have an xy-planar surface, and the display panel **10** may be mounted on the first stage **410**.

[0114] In an embodiment, a first cylinder **420** may be connected below the first stage **410**. The first cylinder **420** may function as a support shaft supporting the first stage **410** and may be connected to a first base plate **430**. The first cylinder **420** may extend in a third direction (for example, a z direction of FIG. 5) and may be arranged between the first stage **410** and the first base plate **430**. According to an embodiment, the first cylinder **420** may be spun or rotated. Here, that the first cylinder **420** is spun may denote that the first cylinder **420** is rotated with respect to a rotational axis in the same direction (the z direction) as the extension direction of the first cylinder **420**, where the rotational axis is positioned in the center of the first cylinder **420**. Here, a driver **600** may be connected to the first cylinder **420**, and the driver **600** may drive the first cylinder **420** to be rotated. The driver **600** may be located in the first cylinder **420**. However, the driver **600** is not limited thereto. The driver **600** may be located outside the first cylinder **420** and may be electrically connected to the first cylinder **420**. The first cylinder **420** may be spun by a certain angle in a plan view to spin and align the first stage **410** connected to the first cylinder **420**. Through this alignment, the first panel and the second panel may be relatively better aligned with each other and bonded to each other.

[0115] In an embodiment and referring to FIGS. 5 and 6, the second support unit **500** may support the second panel, for example, the display circuit board **40**. Hereinafter, an embodiment where the second panel is the display circuit board **40** is mainly described. The display circuit board **40** may be mounted on a second stage **510**. According to an embodiment, the second stage **510** may have a quadrangular cross-section having a long side extending in the first direction (for example, the x direction of FIG. 5) and a short side extending in the second direction (for example, the y direction of FIG. 5) crossing the first direction. However, the second stage **510** may have the short side extending in the first direction and the long side extending in the second direction or may have a square cross-section having the sides having the same length. Accordingly, the second stage **510** may have an xy-planar surface, and the display circuit board **40** may be mounted on the second stage **510**.

[0116] In an embodiment, support portions **520** may be arranged below the second stage **510** to support the second stage **510**. According to an embodiment, the support portions **520** may include a main support portion **520M** and a sub-support portion **520S**.

[0117] In an embodiment, the main support portion **520M** may be arranged below the center of the second stage **510** to support the center of the second stage **510**. An upper end of the main support portion **520M** may be connected to the second stage **510** and a lower end of the main support portion **520M** may be connected to a second base plate **530**. Also, according to an embodiment, the

main support portion **520M** may include a second cylinder **521M**, where the second cylinder **521M** may function as a support shaft supporting the second stage **510** and which may be connected to the second base plate **530**. The second cylinder **521M** may extend in the third direction (for example, the z direction of FIG. 5) and may be arranged between the second stage **510** and the second base plate **530**. According to an embodiment, the second cylinder **521M** may be rotated. It should be appreciated that the fact that the second cylinder **521M** is spun may denote that the second cylinder **521M** is rotated with respect to a rotational axis in the same direction (the z direction) as the extension direction of the second cylinder **521M**, where the rotational axis is positioned in the center of the second cylinder **521M**. According to an embodiment, the driver **600** may be connected to the second cylinder **521M**, and the driver **600** may drive the second cylinder **521M** to be rotated. The driver **600** may be located in the second cylinder **521M**. However, the driver **600** is not limited thereto. In another embodiment, the driver **600** may be located outside the second cylinder **521M** and may be electrically connected to the second cylinder **521M**. The second cylinder **521M** may be rotated by a certain angle in a plan view to spin and align the second stage **510** connected to the second cylinder **521M**. Through this alignment, the second panel and the first panel may be relatively better aligned with each other and bonded to each other.

[0118] In an embodiment, the sub-support portion **520S** may be arranged to support the second stage **510** together with the main support portion **520M**. The sub-support portion **520S** may be arranged at a side of the main support portion **520M**. In detail, the sub-support portion **520S** may be arranged at a side of the main support portion **520M** in a first direction (for example, a +x direction of FIG. 6). Also, the sub-support portion **520** may include two sub-support portions **520**, and in this case, the sub-support portions **520** may be arranged on two sides of the main support portion **520M** in the first direction (for example, the +x direction and a -x direction of FIG. 6). According to another embodiment, the sub-support portion **520S** may include more than two sub-support portions **520**, and in this case, each of the plurality of sub-support portions **520S** may be arranged to be distributed with respect to the main support portion **520M**. For example, when the sub-support portion **520S** includes four sub-support portions **520S**, each of the sub-support portions **520S** may be arranged at an upper side, a lower side, a right side, or a left side with respect to the main support portion **520M** in a plan view. The number of sub-support portions **520S** may be in relation to a size of the second stage **510**. That is, as the size of the second stage **510** increases, an increased number of sub-support portions **520S** may be arranged to support a distributed weight of the second stage **510** together with the main support portion **520M**. Hereinafter, an embodiment where the sub-support portion **520S** includes two sub-support portions as illustrated in FIG. 6 is mainly described.

[0119] According to an embodiment, the sub-support portion **520S** may include a third cylinder **521S** and linear movement portions. The third cylinder **521S** may function as a support shaft supporting the second stage **510**, where the third cylinder **521S** may extend in the third direction (for example, the z direction) and may be arranged between the second stage **510** and the second base plate **530**.

[0120] A first linear movement portion **522S** and a second linear movement portion **523S** may be arranged below the third cylinder **521S**, where the first linear movement portion **522S** may slidably move in the first direction (for example, the x direction of FIG. 6). According to an embodiment, the first linear movement portion **522S** may include a linear motor to slidably move in the first direction. The first linear movement portion **522S** may be fixed to the second base plate **530** there-below. That is, a guide rail of the first linear movement portion **522S** may be fixed to the second base plate **530**, and a movement unit of the first linear movement portion **522S** may slidably move on the guide rail.

[0121] In an embodiment, a connection plate **524S** may be arranged above the first linear movement portion **522S**, that is, above the movement unit of the first linear movement portion **522S**. The connection plate **524S** may be fixed to the movement unit of the first linear movement

portion **522S** and may move together with the movement unit when the movement unit moves in the first direction.

[0122] In an embodiment, the second linear movement portion **523S** may be arranged on the connection plate **524S**, where the second linear movement portion **523S** may slidably move on the connection plate **524S** in a second direction (for example, a y direction of FIG. 6). According to an embodiment, the second linear movement portion **523S** may include a linear motor to slidably move in the second direction. The second linear movement portion **523S** may be fixed to the connection plate **524S** there-below. That is, a guide rail of the second linear movement portion **523S** may be fixed to the connection plate **524S**, and a movement unit of the second linear movement portion **523S** may slidably move on the guide rail.

[0123] In an embodiment, the third cylinder **521S** may be arranged above the second linear movement portion **523S**, that is, above the movement unit of the second linear movement portion **523S**. The third cylinder **521S** may be fixed to the movement unit of the second linear movement portion **523S** and may move together with the movement unit as the movement unit moves in the second direction.

[0124] According to an embodiment, a heater **550** may further be arranged between the main support portion **520M** and the sub-support portion **520S**, and the second stage **510**. The heater **550** may heat the second panel arranged on the second stage **510**, for example, the printed circuit board **40**, to a bonding temperature. Also, a heat insulation layer **560** may further be arranged to be disposed between the heater **550**, and the second cylinder **521M** and the third cylinder **521S**. The heat insulation layer **560** may prevent the transmission of the heat, supplied by the heater **550**, to the second cylinder **521M**, the third cylinder **521S**, and the movement portions below the second cylinder **521M** and the third cylinder **521S**.

[0125] Also, although not shown in the drawings, in an embodiment, an ionizer (not shown) for removing an electrostatic force in a bonding process of the first panel and the second panel may be arranged.

[0126] FIG. 7 is a schematic plan view of the second support unit **500**, according to an embodiment. In FIG. 7, the first support unit **400** is omitted for convenience of explanation and the main support portion **520M** and the sub-support portion **520S** are mainly illustrated.

[0127] In an embodiment and referring to FIG. 7, in a process of bonding the first panel to the second panel, an operation of aligning the second support unit **500** to align the first panel with the second panel, is to be described. As described above, the second panel may be mounted on the second support unit **500** and the pressing unit may perform a pressing operation above the second support unit **500**. Thus, according to the related art, a pressing weight by the pressing unit may be concentrated in the second support unit **500**, and a deflection or pushing problem may particularly occur in the second stage **510** due to the pressing weight. According to an embodiment, the second support unit **500** may include not only the main support portion **520M** but also the at least one sub-support portion **520S**. Thus, the pressing weight may be distributed between the main support portion **520M** and the sub-support portion **520S**, where the main support portion **520M** may be arranged in the center of the second stage **510** and the sub-support portion **520S** may be arranged to be spaced apart from the main support portion **520M** to be disposed adjacent to a circumference of the second stage **510**. Thus, the second stage **510** may be stably supported and the deflection or pushing problem in the second stage **510** may be prevented.

[0128] Also, by including the plurality of support portions, the second support unit **500**, according to an embodiment, may distribute the pressing weight as well as easily correcting the shaft rotation for the alignment of the second panel. In detail, the main support portion **520M** may be rotated by a certain angle by the driver **600**, as described above. Thus, the second stage **510** connected above the main support portion **520M** may be rotated together. In this embodiment, because, in a plan view, the second stage **510** may be rotated with the main support portion **520M** as a rotation axis, the second stage **510** may not be rotated when the sub-support portions **520S**, which are arranged to

be spaced apart from the main support portion **520M** and the rotation axis, are fixed.

[0129] However, according to an embodiment, the sub-support portions **520S** may include the first linear movement portion **522S** slidable in a first direction (an x direction of FIG. 7) and the second linear movement portion **523S** slidable in a second direction (a y direction of FIG. 7). Thus, when the main support portion **520M** is rotated so that the second stage **510** is rotated with the main support portion **520M** as the rotation axis, the sub-support portions **520S** may, with the rotation of the second stage **510**, linearly move in a first direction and a second direction on a plane which is perpendicular to the main support portion **520M** which is the rotation axis. According to an embodiment, the main support portion **520M** and the sub-support portion **520S** may be arranged to be spaced apart from each other by a first distance **D1** in the plan view. Also, when the second stage **510** is rotated by the main support portion **520M**, the sub-support portion **520S** may linearly move in the first direction and the second direction, and thus, the sub-support portion **520S** may move by maintaining the first distance **D1**. In the plan view, the movement of the sub-support portion **520S** may be a linear movement in the first direction and the second direction. However, the movement of the sub-support portion **520S** may be substantially similar to an orbit movement having the main support portion **520M** as a rotation axis.

[0130] As described above, in an embodiment, the main support portion **520M** may be rotated as a fixed end, and the sub-support portion **520S** may be slidingly moved due to the rotation of the second stage **510** as a free end. Therefore, the second support unit **500** may distribute the pressing weight through the plurality of support portions and may easily perform the shaft rotation correction.

[0131] FIG. 8 is a schematic plan view of the second support unit **500**, according to an embodiment and is similar to FIG. 7. The second support unit **500**, according to an embodiment, described with reference to FIG. 8, may be similar to the second support unit **500** described above. Thus, hereinafter, only differences are mainly described.

[0132] In an embodiment and referring to FIG. 8, the sub-support portion **520S** may include a curvature movement portion **525S** disposed between the third cylinder **521S** and the second base plate **530**. That is, the sub-support portion **520S** may include the curvature movement portion **525S** rather than the first linear movement portion **522S** and the second linear movement portion **523S**. The curvature movement portion **525S** may allow the sub-support portion **520S** to perform a curved motion with respect to the main support portion **520M**. In detail, a guide rail of the curvature movement portion **525S** may have a curved shape having a curvature as illustrated in FIG. 8. The guide rail may extend along a portion of an arc having, as a radius, the first distance **D1** between the main support portion **520M** and the sub-support portion **520S**. Here, a movement unit of the curvature movement portion **525S** may move along this curved guide rail. In this case, the connection plate **524S** may be arranged above the curvature movement portion **525S**, that is, above the movement unit of the curvature movement portion **525S**. The connection plate **524S** may be fixed to the movement unit of the curvature movement portion **525S** and may be moved together when the movement unit performs the curved motion. The third cylinder **521S** may be arranged above the connection plate **524S**.

[0133] In an embodiment and as described above, when the main support portion **520M** is rotated so that the second stage **510** is rotated with the main support portion **520M** as a rotation axis, the sub-support portion **520S** connected to the second stage **510** may slidingly move along the guide rail of the curvature movement portion **525S**. Thus, the second support unit **500** may distribute the pressing weight through the plurality of support portions and may easily perform shaft rotation correction.

[0134] In an embodiment, the second support unit **500** according to an embodiment is further described with reference to FIG. 7. The second support unit **500** according to an embodiment may be similar to the second support unit **500** described above. Thus, hereinafter, only differences are mainly described.

[0135] According to an embodiment, the driver **600** may be connected to the sub-support portion **520S**. In detail, the first linear movement portion **522S** and the second linear movement portion **523S** of the sub-support portion **520S** may be driven by the driver **600** to move linearly. Thus, the second stage **510** connected above the third cylinder **521S** of the sub-support portion **520S** may be rotated with the main support portion **520M** as a rotation axis. Here, the main support portion **520M** may not be driven by the driver **600** to be rotated, but may be rotated by the movement of the linear movement portions and the second stage **510**. That is, for comparison, according to the embodiment of FIG. 7, the main support portion **520M** may be driven by the driver **600** to be rotated so that the second stage **510** may be rotated so as to induce the linear movement of the linear movement portions. However, according to an embodiment, the linear movement portions may be driven by the driver **600** to move linearly, so that the second stage **510** may be rotated to induce the spinning of the main support portion **520M**.

[0136] Also, it may be understood that when the second support unit **500** includes the curvature movement portion **525S** rather than the linear movement portions, the driving operation may be similar as described above. That is, the curvature movement portion **525S** may be driven by the driver **600** to perform a curved motion, and thus, the second stage **510** may be rotated with the main support portion **520M** as the rotation axis.

[0137] According to an embodiment, the bonding apparatus **2** for the display apparatus **1** may support the display apparatus **1** by distributing the pressing weight exerted by the pressing unit.

[0138] Accordingly, the panels, for example, the display panel and the display circuit board, may be bonded to each other with increased precision.

[0139] Also, while the pressing weight is distributed on the supported display apparatus, shaft rotation correction of the support units to align the panels may be easily performed.

[0140] Effects of the one or more of the embodiments described above are not limited to the effects described above, and other effects that not are described may be clearly understood by one of ordinary skill in the art.

[0141] It should be understood that embodiments described herein should be considered in a descriptive sense only and not for purposes of limitation. Descriptions of features or aspects within each embodiment should typically be considered as available for other similar features or aspects in other embodiments. While one or more embodiments have been described with reference to the figures, it will be understood by one of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope of the invention. Moreover, the embodiments or parts of the embodiments may be combined in whole or in part without departing from the scope of the invention.

Claims

1. A bonding apparatus for a display apparatus, the bonding apparatus comprising: a first support unit on which a first panel is mounted; and a second support unit on which a second panel which is bonded to the first panel is mounted, wherein the second support unit comprises: a stage on which the second panel is mounted; a main support portion arranged below the stage and supporting the stage, wherein as the main support portion is rotated, the stage is rotated with the main support portion as a rotation axis; a sub-support portion connected below the stage and supporting the stage, wherein as the stage is rotated, the sub-support portion slidingly moves on a plane directed perpendicular to the rotation axis; and a base plate supporting the main support portion and the sub-support portion.
2. The bonding apparatus of claim 1, wherein the main support portion and the sub-support portion are arranged to be spaced apart from each other by a first distance, and while the stage is being rotated, the first distance between the main support portion and the sub-support portion is constant.
3. The bonding apparatus of claim 1, wherein, in a plan view, as the stage is rotated, the sub-

support portion linearly moves in a first direction and a second direction crossing the first direction on the plane.

4. The bonding apparatus of claim 3, wherein the sub-support portion comprises: a first linear movement portion arranged on the base plate and linearly moving in the first direction; a connection plate connected above the first linear movement portion; and a second linear movement portion arranged on the connection plate and linearly moving in the second direction.

5. The bonding apparatus of claim 1, wherein, in a plan view, as the main support portion is rotated, the sub-support portion performs a curve motion with respect to the main support portion.

6. The bonding apparatus of claim 5, wherein the sub-support portion comprises: a curvature movement portion arranged on the base plate and having a curved shape; and a connection plate connected above the curvature movement portion.

7. The bonding apparatus of claim 1, wherein, in a plan view, the sub-support portion is provided in plural and is arranged at each of both sides of the main support portion.

8. The bonding apparatus of claim 1, wherein, in a plan view, the sub-support portion is provided in plural and is arranged at each of an upper side, a lower side, a right side, and a left side of the main support portion.

9. The bonding apparatus of claim 1, wherein the second support unit further comprises a driver driving the main support portion to be rotated, wherein the driver does not drive the sub-support portion.

10. A bonding apparatus for a display apparatus, the bonding apparatus comprising: a first support unit on which a first panel is mounted; and a second support unit on which a second panel which is bonded to the first panel is mounted, wherein the second support unit comprises: a stage on which the second panel is mounted; a main support portion arranged below the stage and supporting the stage; a sub-support portion connected below the stage and arranged to be spaced apart from the main support portion; and a base plate supporting the main support portion and the sub-support portion, wherein as the sub-support portion slidingly moves on a plane perpendicular to a rotation axis of the main support portion, the stage is rotated with respect to the main support portion.

11. The bonding apparatus of claim 10, wherein the main support portion and the sub-support portion are arranged to be spaced apart from each other by a first distance, and while the stage is being rotated, the first distance between the main support portion and the sub-support portion is constant.

12. The bonding apparatus of claim 10, wherein, in a plan view, the stage is rotated as the sub-support portion linearly moves in a first direction and a second direction crossing the first direction on the plane.

13. The bonding apparatus of claim 12, wherein the sub-support portion comprises: a first linear movement portion arranged on the base plate and linearly moving in the first direction; a connection plate connected above the first linear movement portion; and a second linear movement portion arranged on the connection plate and linearly moving in the second direction.

14. The bonding apparatus of claim 10, wherein, in a plan view, as the sub-support portion performs a curve motion with respect to the main support portion, the stage is rotated.

15. The bonding apparatus of claim 14, wherein the sub-support portion comprises: a curvature movement portion arranged on the base plate and having a curved shape; and a connection plate connected above the curvature movement portion.

16. The bonding apparatus of claim 10, wherein, in a plan view, the sub-support portion is provided in plural and is arranged at each of both sides of the main support portion.

17. The bonding apparatus of claim 10, wherein, in a plan view, the sub-support portion is provided in plural and is arranged at each of an upper side, a lower side, a right side, and a left side of the main support portion.

18. The bonding apparatus of claim 10, wherein the second support unit further comprises a driver driving the sub-support portion to linearly move the sub-support portion, wherein the driver does

not drive the main support portion.

19. A method of manufacturing a display apparatus, the method comprising: arranging a first panel on a first support unit; arranging a second panel on a second support unit; correcting a shaft rotation of the second support unit for the alignment of the second panel; and pressing the first panel above the second support unit to bond the first panel to the second panel while a portion of the first panel is disposed on the second support unit to overlap the second panel; wherein the second support unit comprises a stage on which the second panel is mounted, a main support portion arranged below the stage, and a sub-support portion connected below the stage, wherein the correcting of the shaft rotation comprises a step in which the stage is rotated with the main support portion as a rotation axis as the main support portion is rotated, and the sub-support portion slidingly moves on a plane directed perpendicular to the rotation axis as the stage rotated.

20. A method of manufacturing a electronic device, the method comprising: manufacturing a display apparatus; and accommodating the display apparatus in the housing, wherein manufacturing the display apparatus comprises arranging a first panel on a first support unit; arranging a second panel on a second support unit; correcting a shaft rotation of the second support unit for the alignment of the second panel; and pressing the first panel above the second support unit to bond the first panel to the second panel while a portion of the first panel is disposed on the second support unit to overlap the second panel; wherein the second support unit comprises a stage on which the second panel is mounted, a main support portion arranged below the stage, and a sub-support portion connected below the stage, wherein the correcting of the shaft rotation comprises a step in which the stage is rotated with the main support portion as a rotation axis as the main support portion is rotated, and the sub-support portion slidingly moves on a plane directed perpendicular to the rotation axis as the stage rotated.
